Física II –Aula 17

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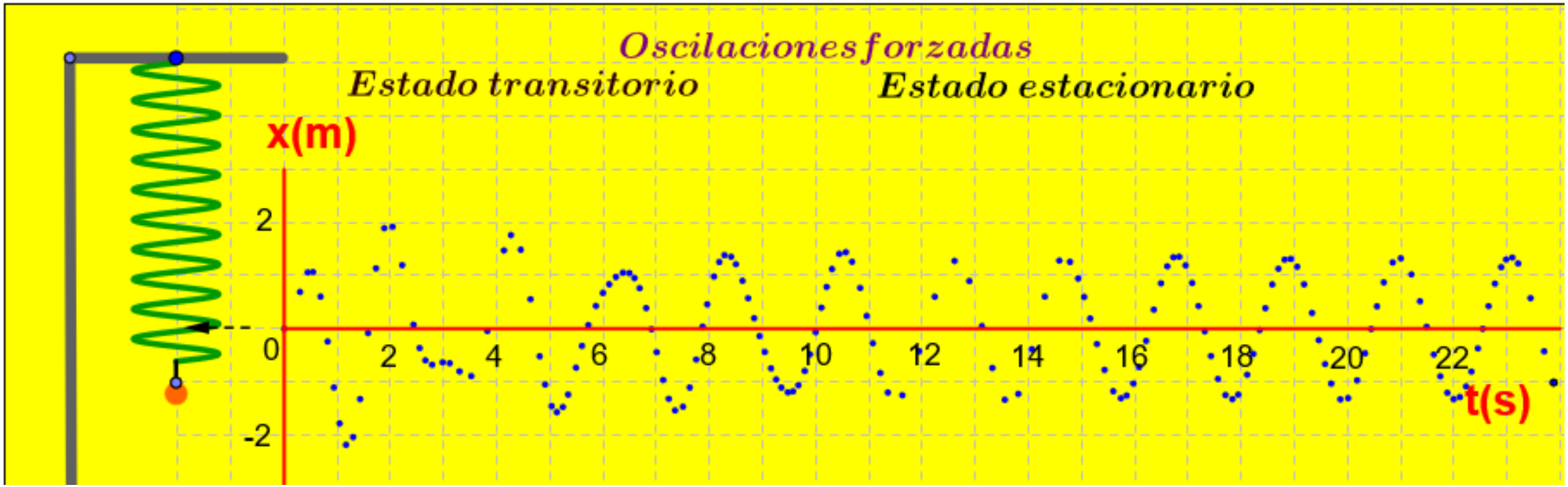
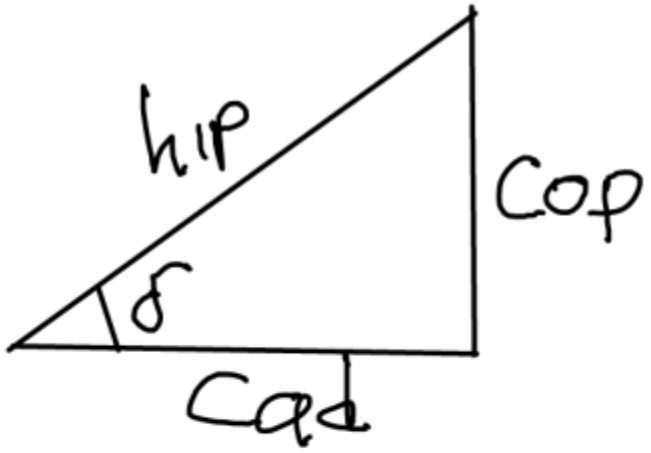


$$x(t) = A e^{-\xi \omega_0 t} \sin(\omega t + \varphi) + A^* \sin(\Omega t + \delta)$$

$$A^* = \frac{-q}{b\Omega}\sin(\delta)$$

$$\tan\delta = -\frac{2\xi\omega_0\Omega}{\omega_0^2-\Omega^2}$$

$$\sin\delta = \frac{cop}{hip}$$



$$\tan\delta = \frac{cop}{cad}$$

$$\sin\delta = \frac{cop}{\sqrt{cop^2 + cad^2}}$$

$$\sin\delta = \frac{-2\xi\omega_0\Omega}{\sqrt{4\xi^2\omega_0^2\Omega^2 + (\omega_0^2-\Omega^2)^2}}$$

$$\sin\delta = \frac{Cop}{\sqrt{Cop^2 + cad^2}}$$

$$\boxed{A^* = -\frac{q}{b\Omega}\sin(\delta)}$$

$$\sin\delta = \frac{-b\Omega}{\pm\sqrt{(K-m\Omega^2)^2 + b^2\Omega^2}} = \frac{-2\xi\Omega\omega_0}{\pm\sqrt{(\omega_0^2-\Omega^2)^2 + 4\xi^2\Omega^2\omega_0^2}}$$

$$\boxed{A^* = \frac{q/m}{\sqrt{(\omega_0^2-\Omega^2)^2 + 4\xi^2\Omega^2\omega_0^2}}}$$

Amplitude de ressonância

$A^*_{res} \rightarrow$ Denominador de A^* seja mínimo

$$\frac{d}{d\Omega}\left(\underbrace{(\omega_0^2 - \Omega^2)^2} + 4\xi^2\Omega^2\omega_0^2 \right) = 2(\omega_0^2 - \Omega^2)(-2\Omega) + 8\xi^2\omega_0^2\Omega = 0$$

$$= 4\Omega\left[(\omega_0^2 - \Omega^2) - 2\xi^2\omega_0^2 \right] = 0$$

$$(\omega_0^2 - \Omega^2) - 2\xi^2\omega_0^2 = 0$$

$$\sqrt{\omega_0^2 - 2\xi^2\omega_0^2} = \Omega_r$$

$$\boxed{\omega_0\sqrt{1-2\xi^2} = \Omega_r}$$

$$\Omega_r = \sqrt{\frac{K}{m} - \frac{b^2}{2m^2}}$$

$$A_r^* = \frac{q}{b\omega_0\sqrt{1-\xi^2}}$$

$$A^* = \frac{q/m}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\xi^2\Omega^2\omega_0^2}}$$

$$\boxed{\Omega_r = \sqrt{\frac{k}{m} - \frac{b^2}{2m^2}} = \omega_0\sqrt{1 - 2\xi^2}.}$$

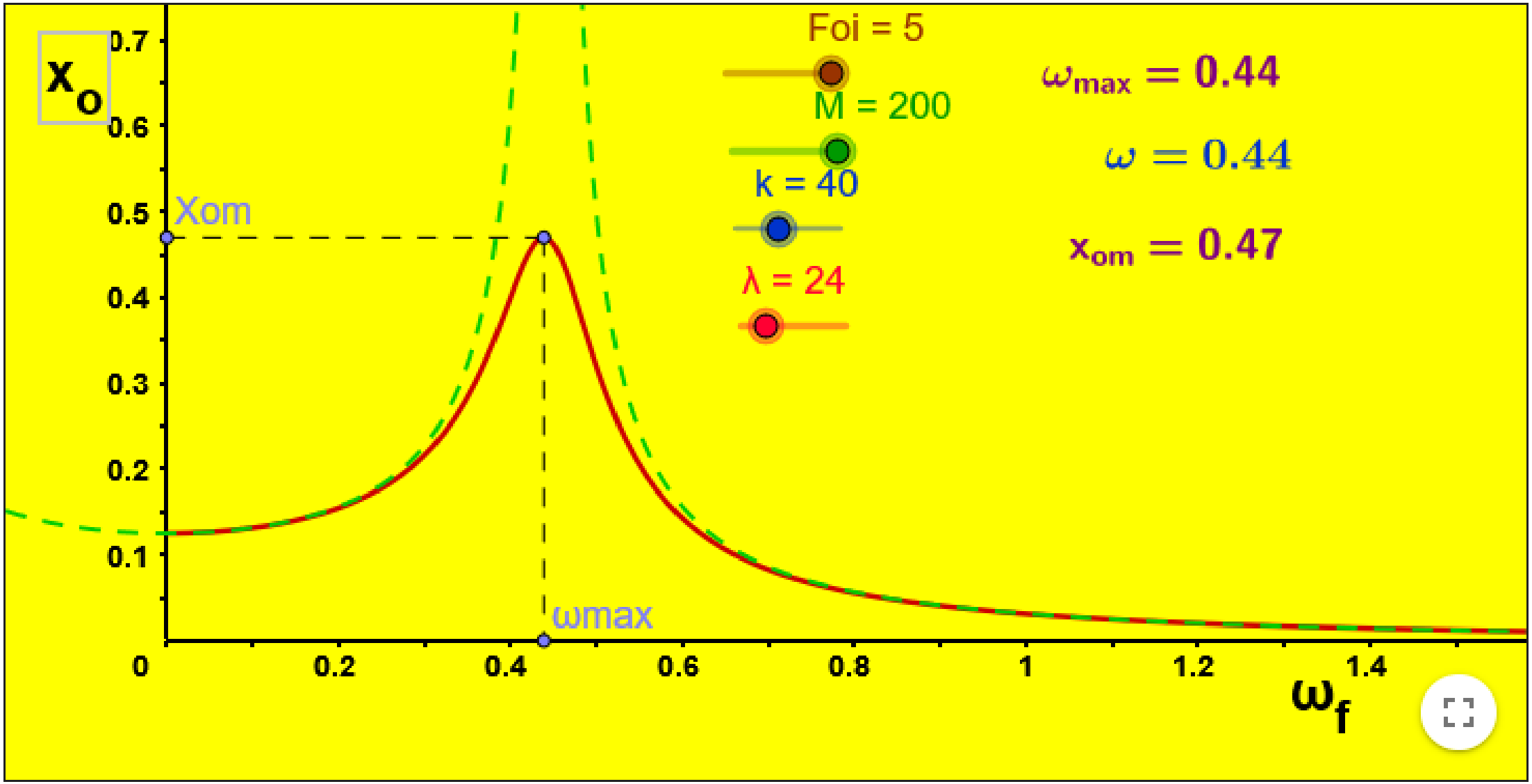
$$\Omega_r^2 = \omega_0^2 - 2\omega_0^2\xi^2; \quad (\Omega_r^2 - \omega_0^2)^2 = 4\omega_0^4\xi^4$$

$$4\xi^2\omega_0^2\Omega_r^2 = 4\xi^2\omega_0^2(\omega_0^2 - 2\omega_0^2\xi^2) = 4\xi^2\omega_0^4 - 8\xi^4\omega_0^4$$

$$(\Omega_r^2 - \omega_0^2)^2 + 4\xi^2\omega_0^2\Omega_r^2 = 4\omega_0^4\xi^4 + 4\xi^2\omega_0^4 - 8\xi^4\omega_0^4$$

$$= 4\xi^2\omega_0^4(1 - \xi^2)$$

$$A_r^* = \frac{q/m}{2\xi\omega_0^2\sqrt{1-\xi^2}} \xRightarrow{b = 2m\omega_0\xi} A_r^* = \frac{q}{b\omega_0\sqrt{1-\xi^2}}$$


x_o
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.2
0.4
0.6
0.8
1
1.2
1.4
ω_f
Xom
ωmax
Foi = 5
M = 200
k = 40
λ = 24
$\omega_{max} = 0.44$
$\omega = 0.44$
$x_{om} = 0.47$

$$\omega_0^2 = \frac{k}{m} \qquad \gamma = \frac{b}{m} \qquad \alpha_0 = \frac{F_0}{m}$$

$$x = A^* \operatorname{sen}(\omega t + \delta) \qquad \dot{x} = \omega A^* \cos(\omega t + \delta)$$

$$v_{\text{máx}} = \omega A^* = \frac{\alpha_0 \omega}{[(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2]^{1/2}}$$

$$v_{\text{máx}} = \frac{F_0}{\sqrt{b^2 + (m\omega - k/\omega)^2}}$$

Impedância

$$Z = \sqrt{b^2 + \left(m\omega - \frac{k}{\omega}\right)^2}$$

Resistência

$$R \equiv b$$

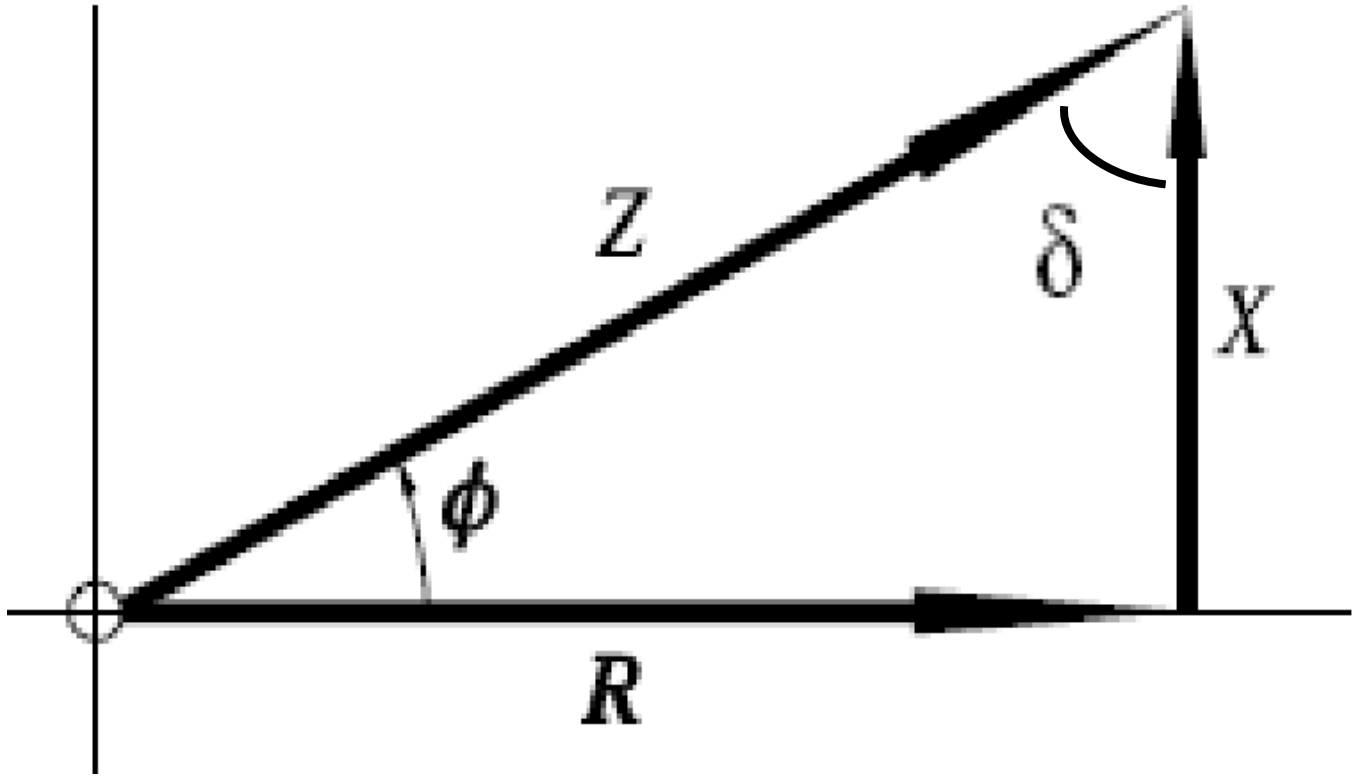
Reatância

$$X \equiv \left(m\omega - \frac{k}{\omega}\right)$$

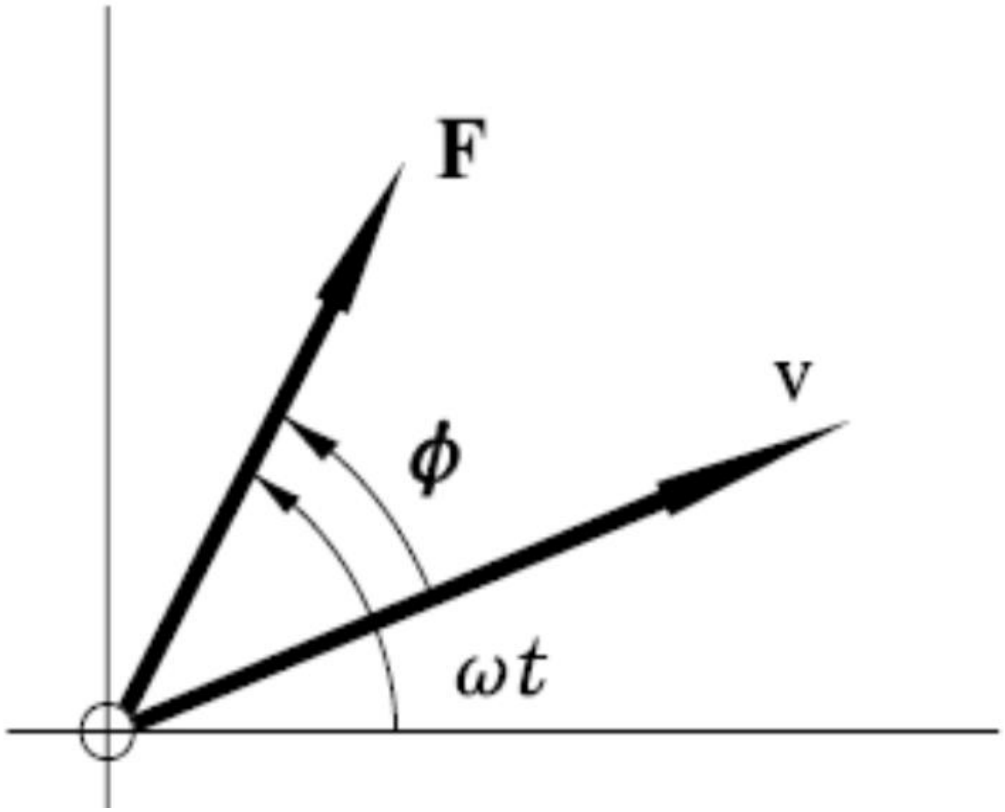
$$\mathrm{tg}\delta = \frac{b}{m\omega - \dfrac{k}{\omega}}$$

$$R \equiv b \qquad X \equiv \left(m\omega - \frac{k}{\omega}\right)$$

$$Z = \sqrt{R^2 + X^2}$$



$$\mathrm{tg}\ \delta = \mathrm{cotg}\ \phi = 1/\mathrm{tg}\ \phi$$



$$\dot{x} = \omega A^{*} \cos(\omega t + \delta)$$

$$tg(\delta) = -tg(-\delta) = cotg(\phi)$$

$$v = \frac{F_0}{Z} \operatorname{sen}(\omega t - \phi)$$

$$P = F v = \frac{F_0^2}{Z} \operatorname{sen}\omega t \ \operatorname{sen}(\omega t - \phi)$$

$$P = \frac{F_0^2}{Z} \left(\operatorname{sen}^2\omega t \cos\phi - \operatorname{sen}\omega t \cos\omega t \operatorname{sen}\phi \right)$$

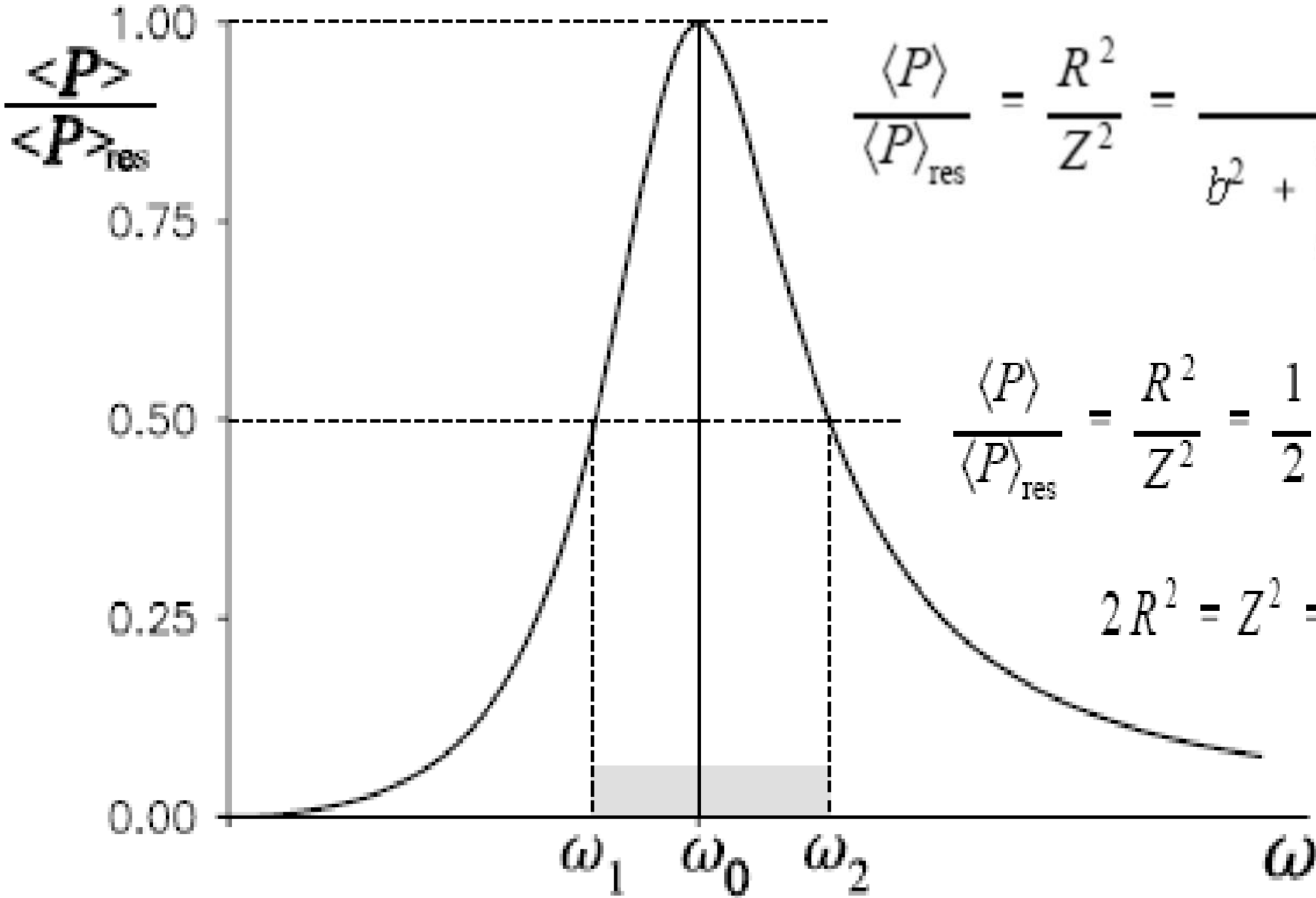
$$\int \sin^2 x\, dx = \frac{1}{2}x - \frac{1}{4}\sin(2x).$$

$$\langle P \rangle = \frac{1}{2}\frac{F_0^2}{Z}\cos\phi = \frac{1}{2}\frac{F_0^2}{Z}\frac{R}{Z} = \frac{1}{2}\frac{F_0^2 R}{Z^2}$$

$\phi=0$ $\omega=\omega_0$

$$\langle P \rangle_{\text{res}} = \frac{1}{2}\frac{F_0^2}{R}$$

$$\frac{\langle P \rangle}{\langle P \rangle_{\text{res}}} = \frac{R^2}{Z^2} = \frac{b^2}{b^2 + \left(m\omega - \frac{k}{\omega}\right)^2}$$



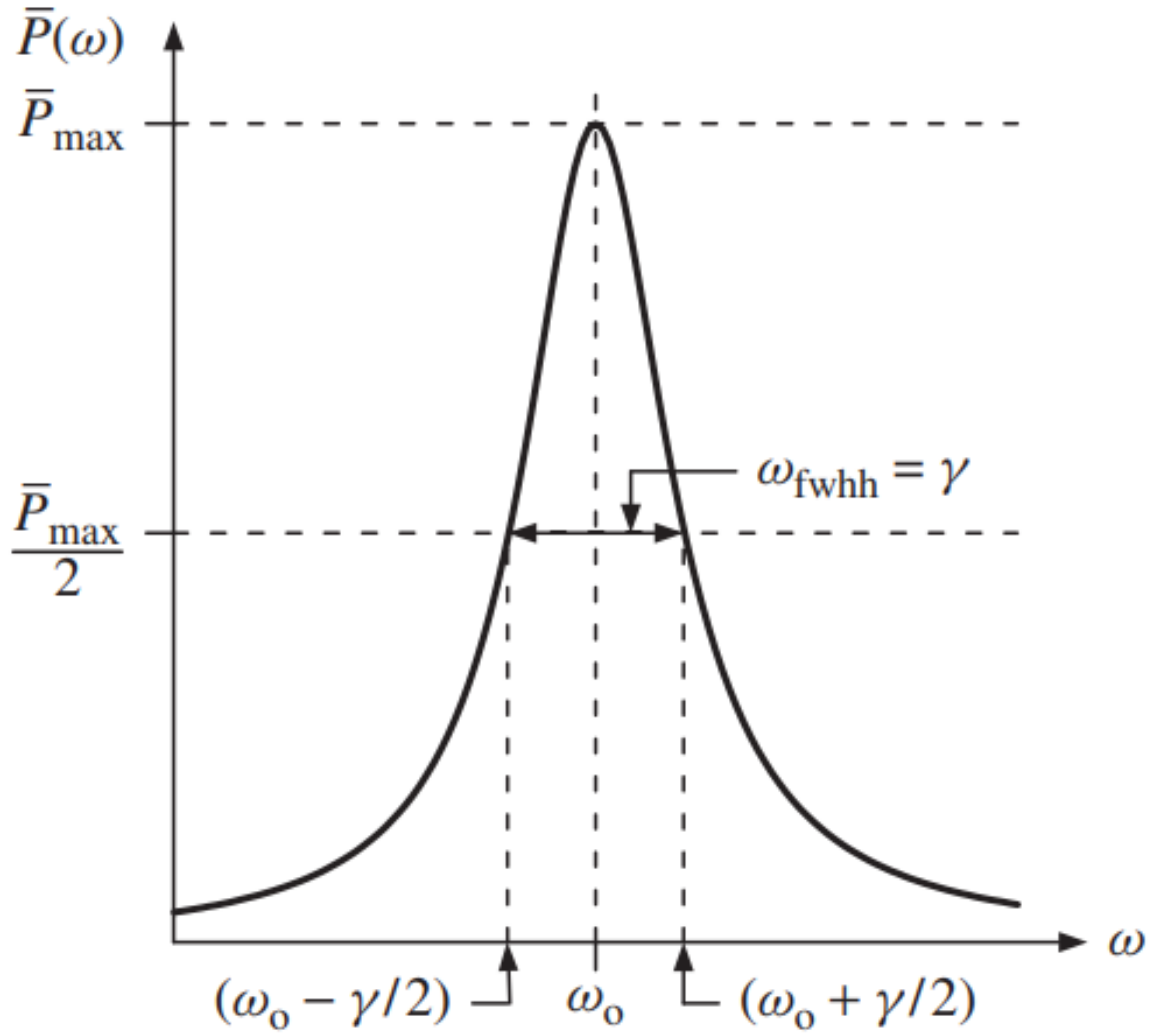
$$\frac{\langle P \rangle}{\langle P \rangle_{res}} = \frac{R^2}{Z^2} = \frac{b^2}{b^2 + \left(m\omega - \frac{k}{\omega}\right)^2}$$

$$\frac{\langle P \rangle}{\langle P \rangle_{res}} = \frac{R^2}{Z^2} = \frac{1}{2}$$

$$2R^2 = Z^2 = R^2 + X^2$$

$$R = \pm X$$

$$\langle P \rangle = \frac{1}{2}\frac{F_0^2}{Z}\cos\phi \quad \Longrightarrow \quad \overline{P}(\omega) = \frac{\omega^2 F_0^2 \gamma}{2m[(\omega_o^2 - \omega^2)^2 + \omega^2\gamma^2]}.$$

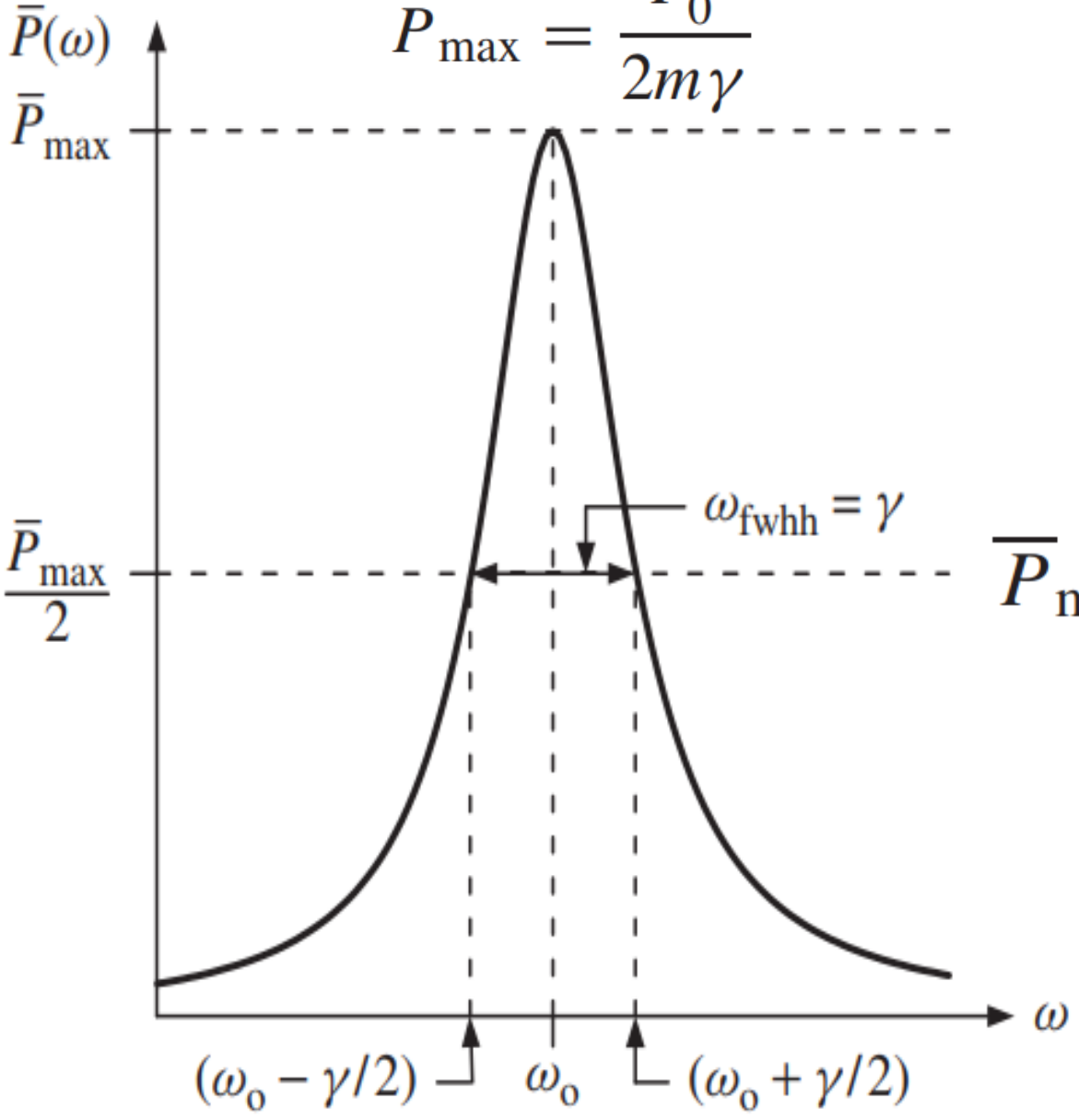


$$(\omega^2 - \omega_o^2) = (\omega_o + \omega)(\omega_o - \omega) \approx 2\omega_o(-\Delta\omega),$$

$$\omega \approx \omega_o$$

$$\overline{P}(\omega) = \frac{\omega^2 F_0^2 \gamma}{2m[(\omega_o^2 - \omega^2)^2 + \omega^2\gamma^2]}. \quad \Rightarrow \quad \boxed{\overline{P}(\omega) = \frac{F_0^2}{2m\gamma(4\Delta\omega^2/\gamma^2 + 1)}.}$$

$$\Delta\omega = 0 \quad \Rightarrow \quad \overline{P}_{\max} = \frac{F_0^2}{2m\gamma}$$



$$\overline{P}_{max} = \frac{F_0^2}{2m\gamma}$$

$$\boxed{\overline{P}(\omega) = \frac{F_0^2}{2m\gamma(4\Delta\omega^2/\gamma^2 + 1)}.}$$

$$\overline{P}_{max}/2 \longrightarrow 2\Delta\omega/\gamma = 1$$

$$2\Delta\omega = \gamma$$

$$\boxed{\omega_{fwhh} = 2\Delta\omega = \gamma = \omega_o/Q}$$

Exemplo

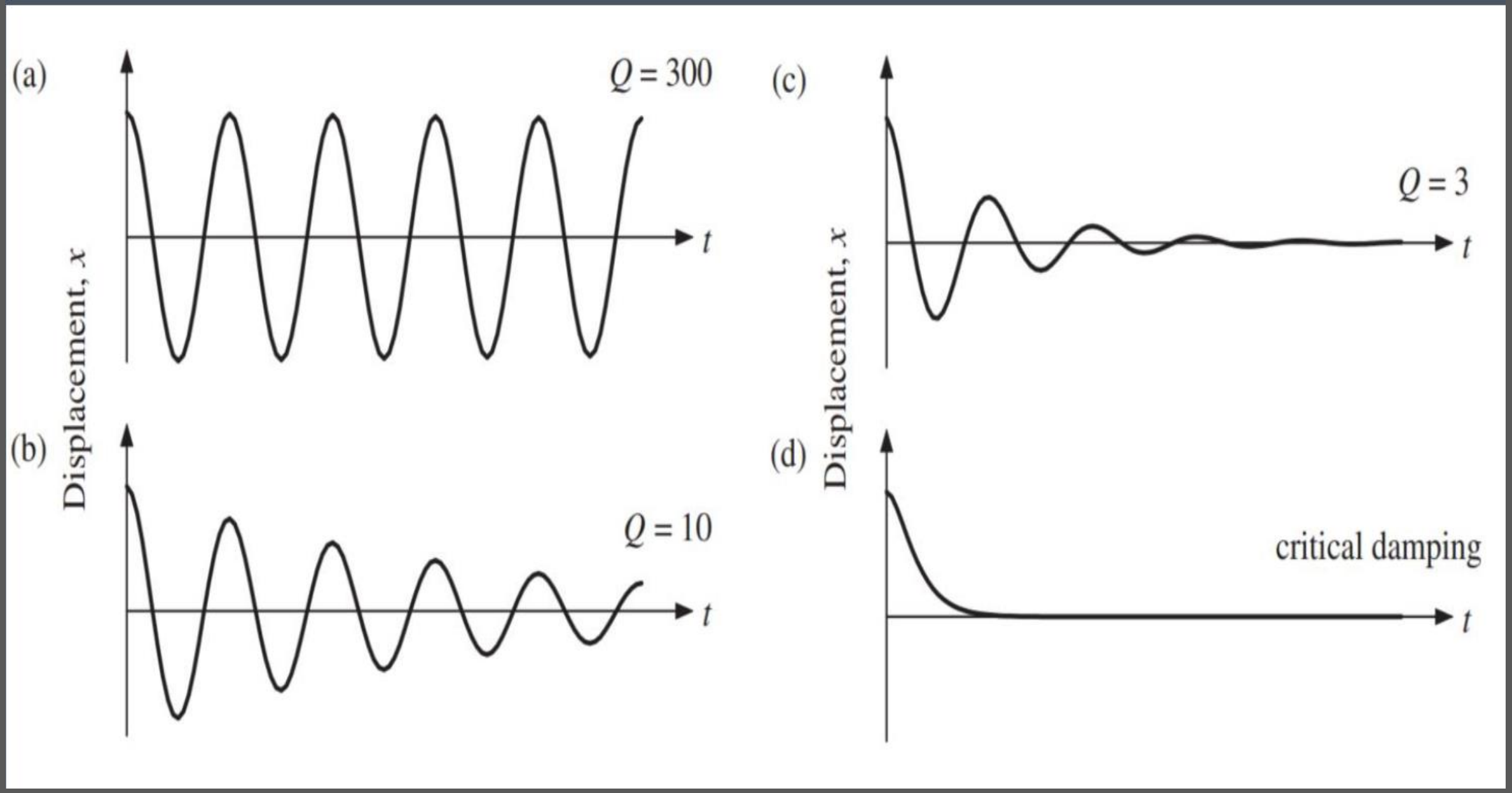
- Um objeto de massa 1,5 kg em uma mola de constante de elasticidade 600 N/m perde 3 por cento de sua energia em cada ciclo. O mesmo sistema é impulsionado por uma força senoidal com valor máximo Fmax = 0,5 N. (a) Qual é Q para este sistema? (b) Qual é a frequência de ressonância (angular)? (c) Se a frequência varia lentamente através da ressonância, qual é a largura Deltaw da ressonância? (d) Qual é a amplitude na ressonância? (e) Qual é a amplitude se a frequência de acionamento é w = 19 rad / s?

(a) Quál e o valor de Q para este sistema?

$$Q \approx \frac{2\pi}{(|\Delta E|/E)_{\text{cycle}}} = \frac{2\pi}{0.03} = \boxed{209}$$

(b) Qual é a frequência de ressonância (angular)?

$$\omega_0 = \sqrt{\frac{k}{m}} = \boxed{20\ \text{rad/s}}$$


(a)
Q = 300
t
Displacement, x
(b)
Q = 10
t
(c)
Q = 3
t
Displacement, x
(d)
critical damping
t

(c) Se a frequência varia lentamente através da ressonância, qual é a largura de banda da ressonância?

$$\omega_{\text{fwhh}} = \frac{\omega_0}{Q} = \boxed{0.0957 \text{ rad/s}}$$

(d) Qual é a amplitude em w=w0?

$$A(\omega) = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + b^2\omega^2}}$$

$$A(\omega_0) = \frac{F_0}{b\omega_0}$$

$$b = \frac{m\omega_0}{Q} = 0.144 \text{ kg/s}$$

$$A(\omega_0) = \frac{F_0}{b\omega_0} = \boxed{17.4 \text{ cm}}$$

(e) Qual é a amplitude se a frequência de acionamento é w = 19 rad / s?

$$A(19 \text{ s}^{-1}) = \frac{0.5}{\sqrt{1.5^2(20^2 - 19^2)^2 + 0.144^2(19)^2}}$$

$$= \boxed{0.854 \text{ cm}}$$

Exemplo

- Uma massa m é suspensa em equilíbrio por uma mola que exerce uma força F = -K (x -l), onde *x* é o comprimento da mola e l é seu comprimento quando está relaxada. Em t = 0, o ponto de suporte ao qual a extremidade superior da mola está fixada começa oscilar sinusoidalmente para cima e para baixo com amplitude A, frequência angular *w*, conforme mostrado na Fig. 1. Configure e resolva a equação de movimento para *x(t)*.

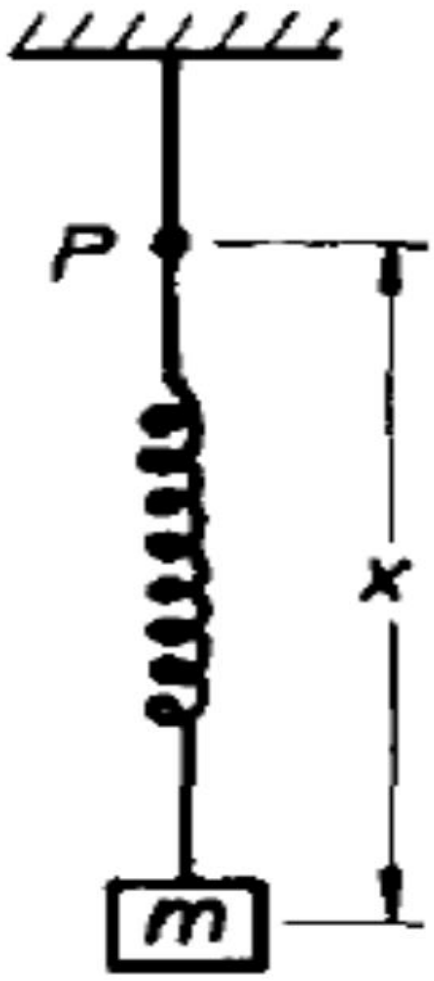


Fig. 1.

$$m\frac{d^2}{dt^2}[x + A\sin(\omega t)] = mg - K(x - l)$$

$$y = x - l - \frac{mg}{K},\ \omega_0^2 = \frac{K}{m}$$

$$\ddot{y} + \omega_0^2 y = \omega^2 A \sin(\omega t)$$

particular solution $y = B\sin(\omega t)$

$$B = \frac{\omega^2 A}{\omega_0^2 - \omega^2}$$

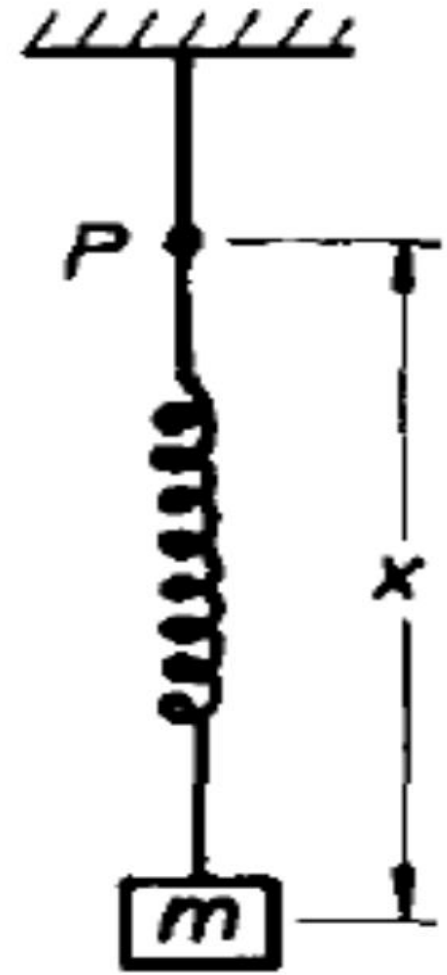


Fig. 1.

$$y = C\cos(\omega_0 t) + D\sin(\omega_0 t) + \frac{\omega^2 A \sin(\omega t)}{\omega_0^2 - \omega^2}$$

$$\dot{y} = -C\omega_0 \sin(\omega_0 t) + D\omega_0 \cos(\omega_0 t) + \frac{\omega^3 A \cos(\omega t)}{\omega_0^2 - \omega^2}$$

$$mg = K(x - l), \text{ i.e. } x = \frac{mg}{K} + l, \text{ or } y = 0$$

$$\dot{y} = 0$$

$$C = 0, \qquad D = \frac{-\omega^3 A}{\omega_0(\omega_0^2 - \omega^2)}$$

$$x(t) = \frac{\omega^2 A}{\omega_0^2 - \omega^2}\left[\sin(\omega t) - \frac{\omega}{\omega_0}\sin(\omega_0 t)\right] + \frac{mg}{K}$$

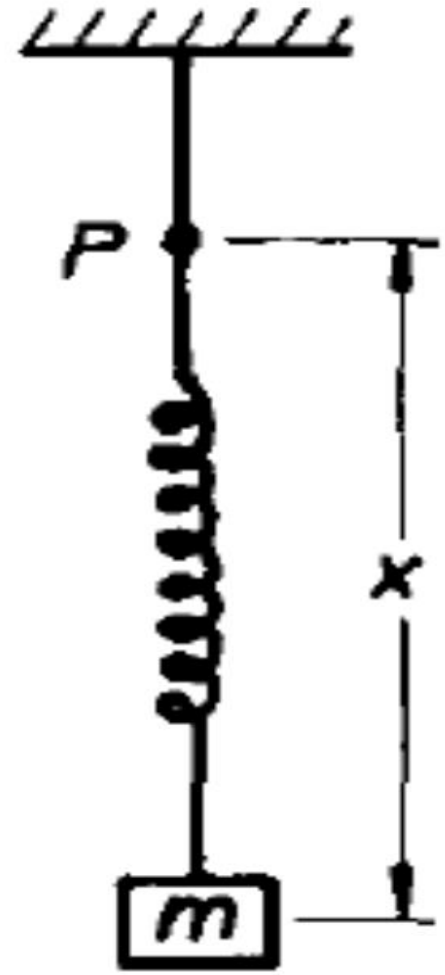


Fig. 1.

Exemplo

- Um carro está viajando na direção *x*, e mantém a velocidade horizontal *v* constante. O carro passa por uma lombada cuja forma é descrita por $y_0 = A\,[1 - \cos \pi x / l)]$ para $0 \leq x \leq 2l$; e $y_0 = 0$ caso contrário (Fig. 3). Determine o movimento do centro de massa do carro ao passar sobre a lombada. Represente o carro como uma massa *m,* presa a uma mola sem massa de comprimento natural l_0 e constante de mola *k*. Ignore o atrito e presuma que a mola está sempre na vertical.

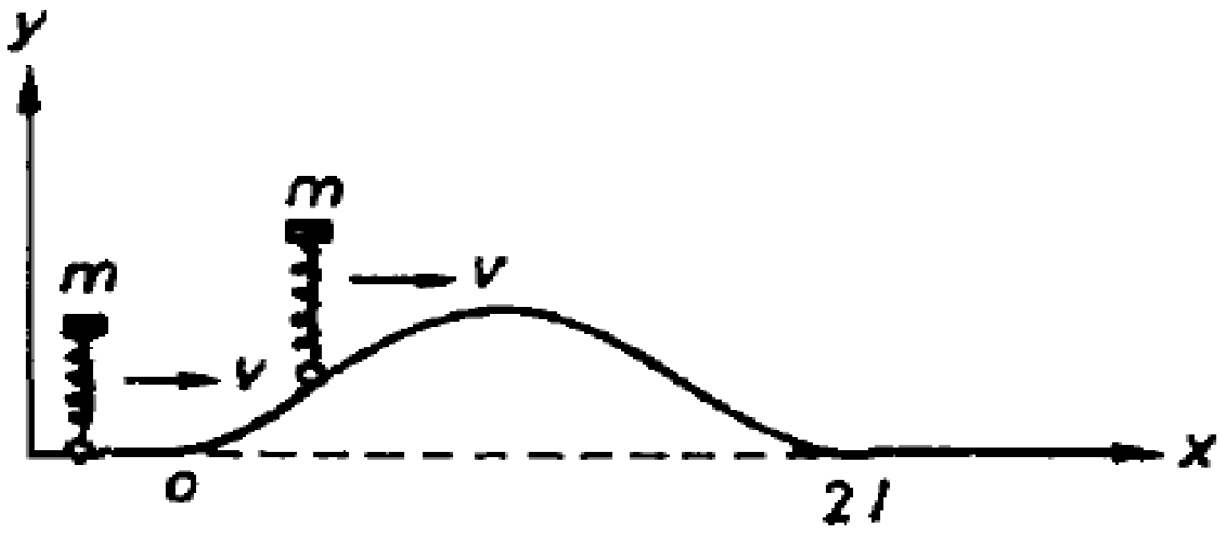
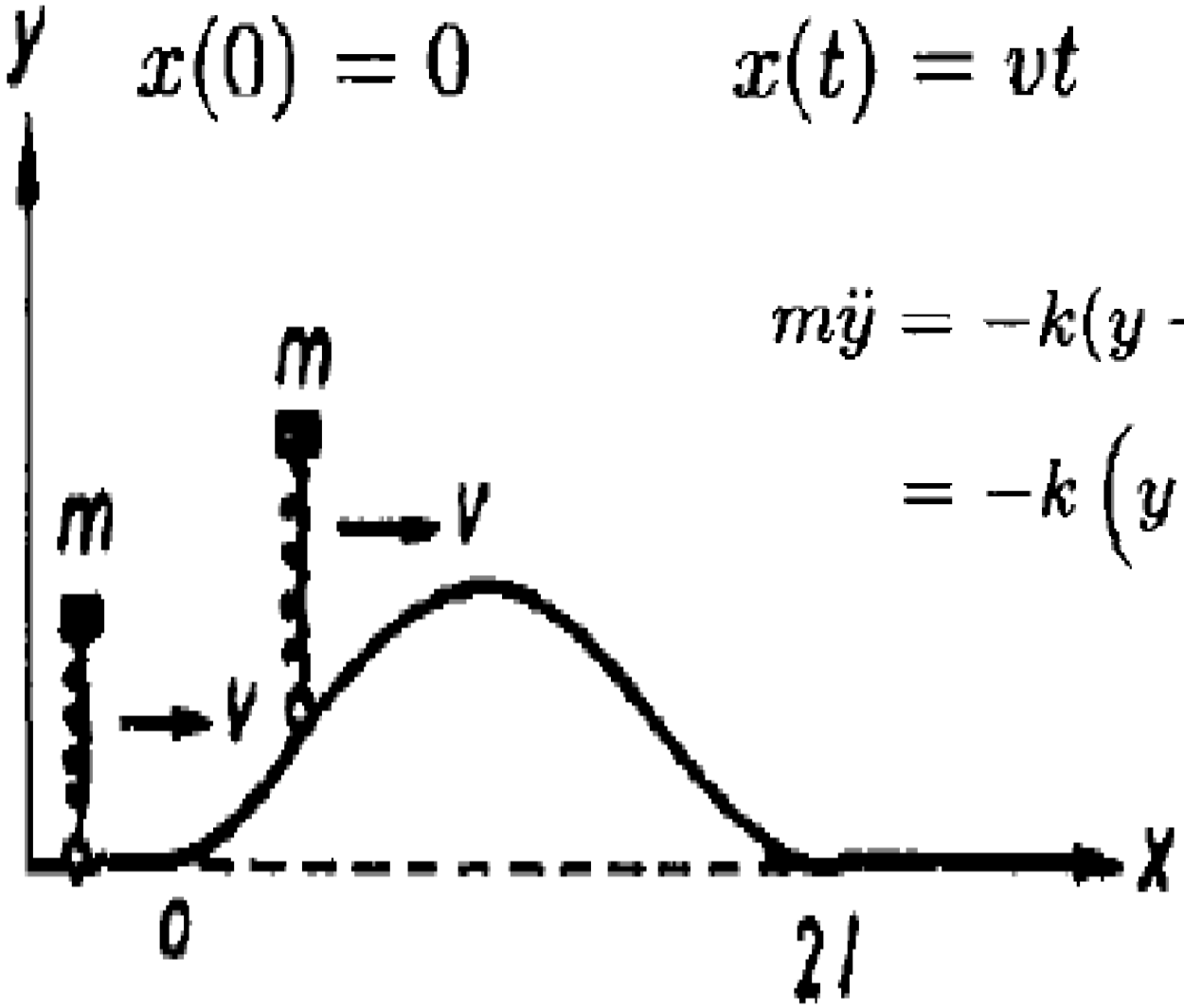


Fig. 3.

$y_0 = A\,[1 - \cos \pi x/ l)]$

$x(0) = 0 \qquad x(t) = vt$



$$m\ddot{y} = -k(y - y_0 - l_0) - mg$$

$$= -k\left(y - A - l_0 + \frac{mg}{k}\right) - kA\cos\left(\frac{\pi vt}{l}\right)$$

$$Y = y - A - l_0 + mg/k$$

Fig. 3.

$$m\ddot{Y} + kY = -kA\cos\left(\frac{\pi v t}{l}\right)$$

particular solution $\quad Y = B\cos(\frac{\pi v t}{l})$

$$-mB\left(\frac{\pi v}{l}\right)^2 + kB = -kA$$

$$B = \frac{kl^2 A}{m\pi^2 v^2 - kl^2}$$

$$y = C_1 \cos(\omega t) + C_2 \sin(\omega t) + B \cos\left(\frac{\pi v t}{l}\right) + A + l_0 - \frac{mg}{k}$$

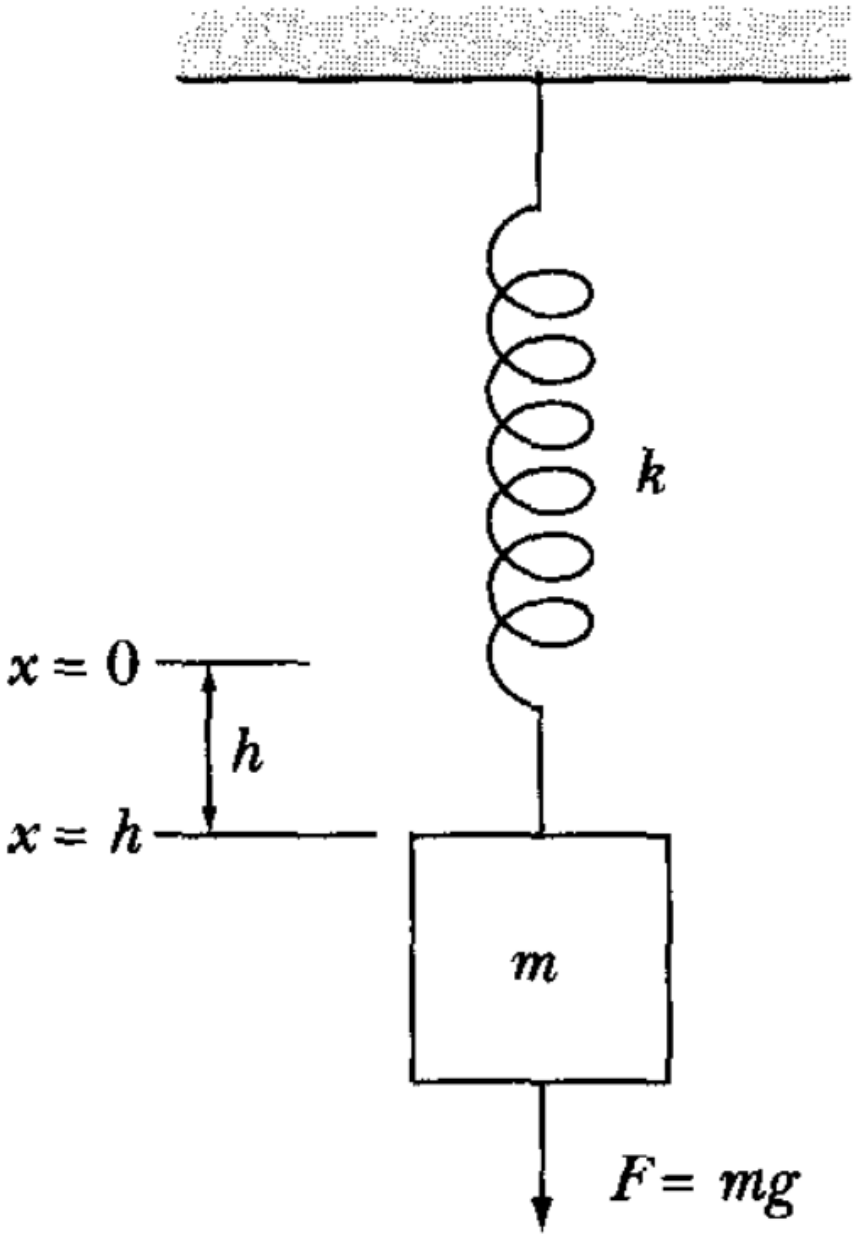
$$\omega = \sqrt{\frac{k}{m}}.$$

$$y(0) = l_0 - mg/k \qquad \dot{y}(0) = 0$$

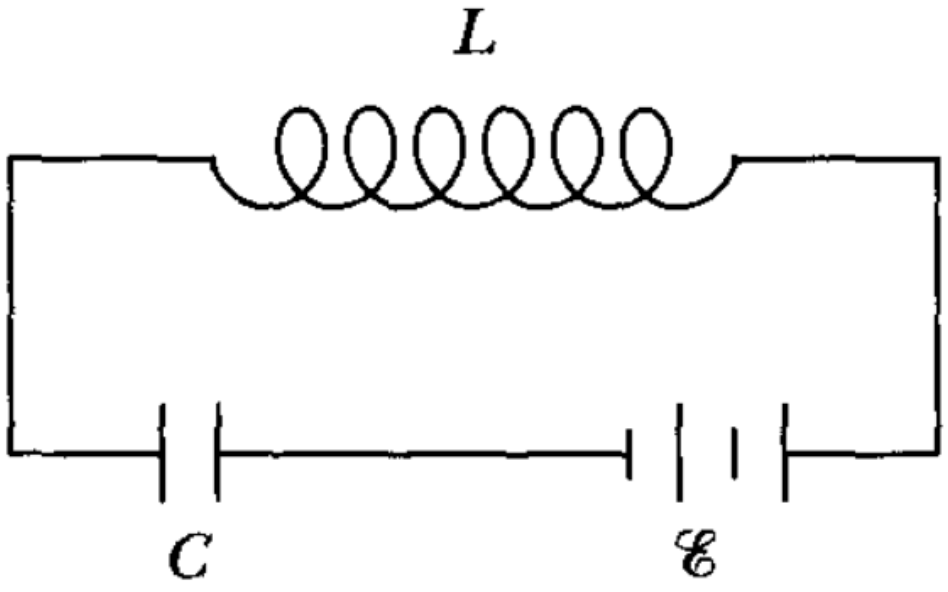
$$C_2 = 0 \qquad C_1 = -(B + A) = m\pi^2 v^2 A/(kl^2 - m\pi^2 v^2)$$

$$y(t) = C_1 \cos(\omega t) + B \cos\left(\frac{\pi v t}{l}\right) + A + l_0 - \frac{mg}{k}$$

$$\omega = \sqrt{\frac{k}{m}},\ C_1 = -\frac{m\pi^2 v^2 A}{m\pi^2 v^2 - kl^2},\ B = \frac{kl^2 A}{m\pi^2 v^2 - kl^2}$$



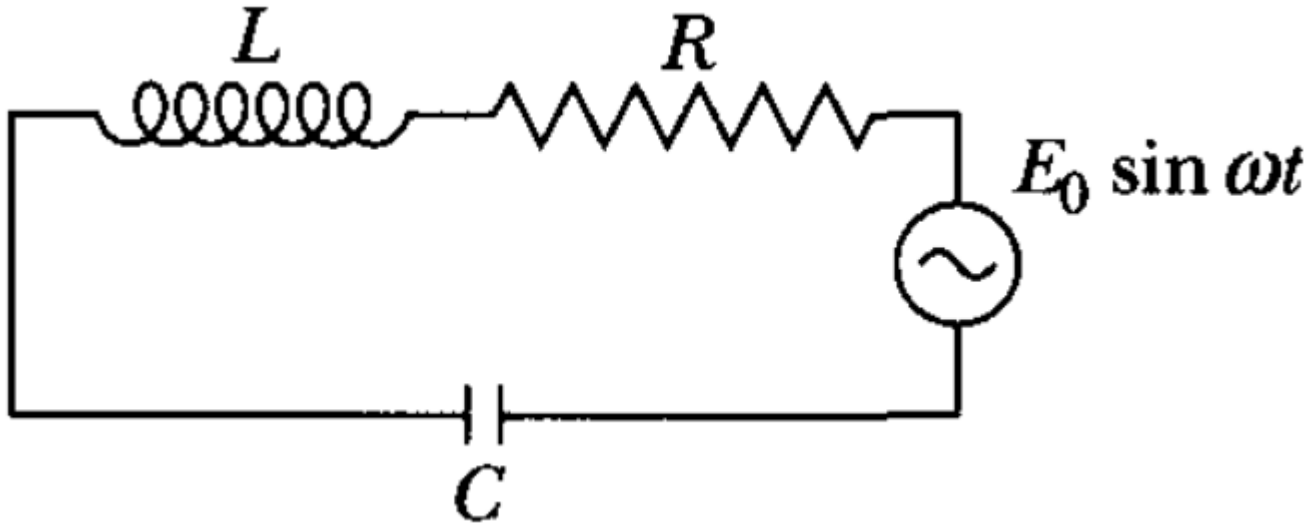
(a)



(b)

Mechanical		Electrical	
x	Displacement	q	Charge
$\dot{x}$	Velocity	$\dot{q} = I$	Current
m	Mass	L	Inductance
b	Damping resistance	R	Resistance
$1/k$	Mechanical compliance	C	Capacitance
F	Amplitude of impressed force	$\mathcal{E}$	Amplitude of impressed emf

$$V_L = L\frac{dI}{dt} = L\ddot{q}$$



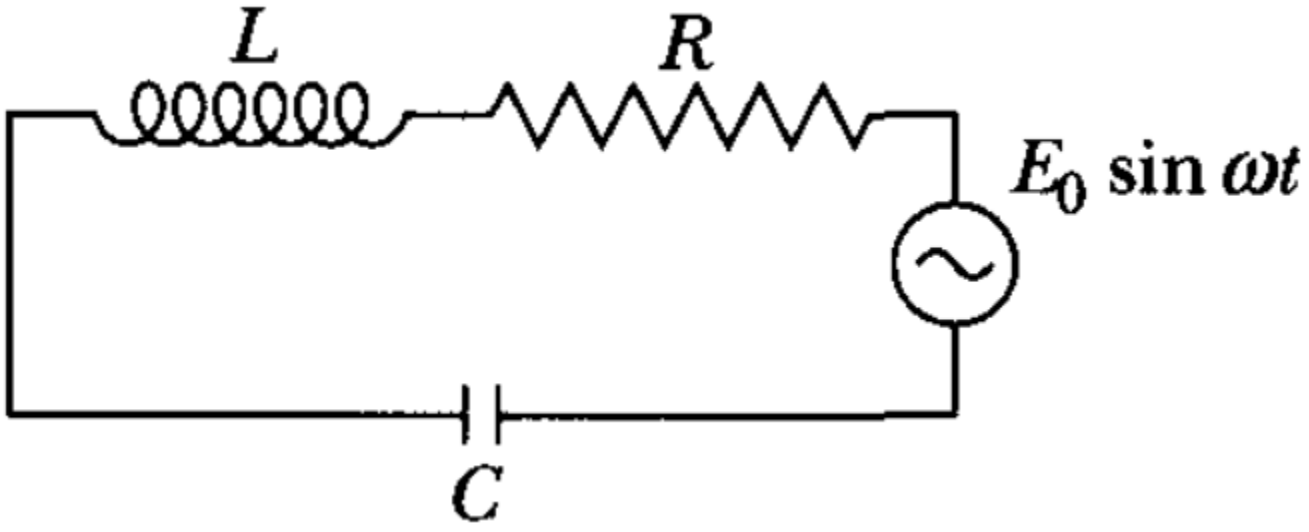
$$V_R = RI = R\frac{dq}{dt} = R\dot{q}$$

$$V_C = \frac{q}{C}$$

$$L\ddot{q} + R\dot{q} + \frac{q}{C} = E_0 \sin \omega t$$

$$\omega_0 = \sqrt{k/m} \rightarrow 1/\sqrt{LC}$$

$$A = F_0/m \rightarrow E_0/L.$$



$$\dot{x} = \frac{-A\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \sin(\omega t + \delta)$$

$$I = \frac{-E_0}{\sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}} \sin(\omega t + \delta)$$

$$V_L = L\frac{dI}{dt} = \frac{-\omega L E_0}{\sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}} \cos(\omega t - \delta)$$

$$= V(\omega)\cos(\omega t - \delta)$$

ω_{max}

$$\frac{dV(\omega)}{d\omega} = \frac{LE_0\left(R^2 - \frac{2L}{C} + \frac{2}{\omega^2 C^2}\right)}{\left[R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2\right]^{3/2}}$$

Fazendo o numerador igual a zero

$$\omega_{\max} = \frac{1}{\sqrt{LC - \frac{R^2C^2}{2}}}$$

$$\omega_0 = 1/\sqrt{LC}$$

$$\omega_R = \sqrt{1/LC - 2R^2/L^2}$$